[CONTROLLING THE FACTORS THAT AFFECT THE SCREED]

The screed will float at the same position as long as all factors that affect the screed remain unchanged.

A floating screed is towed by the tractor portion of the paver and supported by the bituminous material that passes under the screed nose. When the paver takes off from the starting reference, reaches the planned speed and develops uniform operation of the material feed system, the screed will float at that position unless one of the factors is changed.

Paving Speed. As described in the previous section, the speed of the paver affects how the screed floats. The screed can reach its planing angle and its equilibrium point at any speed. As long as the speed remains the same, the screed maintains its vertical position. When the speed changes, the screed reacts because the shear factor also changes.

Factors that affect the screed:

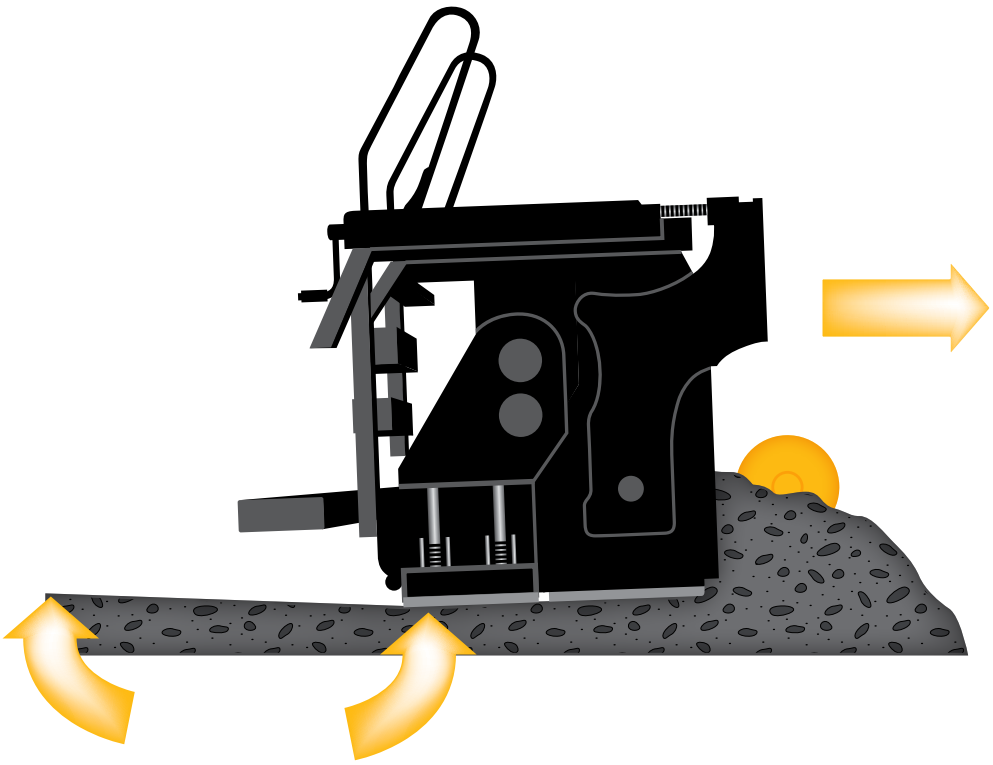
1. **Paving Speed**
2. **Head of Material**
3. **Screed Adjustments**
4. **Type of Mix**
5. **Temperature of Mix**
6. **Grade Conditions**



User Tip: *A common time for the operator to increase the paving speed is when haul units begin to accumulate in front of the paver. Or, the operator may increase the paving speed at the end of a pull when the last truck is being paved out. Caterpillar recommends paving at a continuous, calculated speed with the understanding that there may be times when several haul units accumulate in front of the paver. Resist the temptation to pave faster. High-speed paving will not increase production. The paver will simply stop more often waiting for mix, and quality will suffer.*

Increasing the paving speed will cause the screed to drop.

When the paving speed is increased, more energy is available to push the head of material in front of the screed. Less material flows under the nose of the screed and layer thickness decreases. If the automatic grade control system is in use, the system will detect the change in layer thickness and call for a correction. Still, there will be a dip in the bituminous layer, resulting in increased roughness.

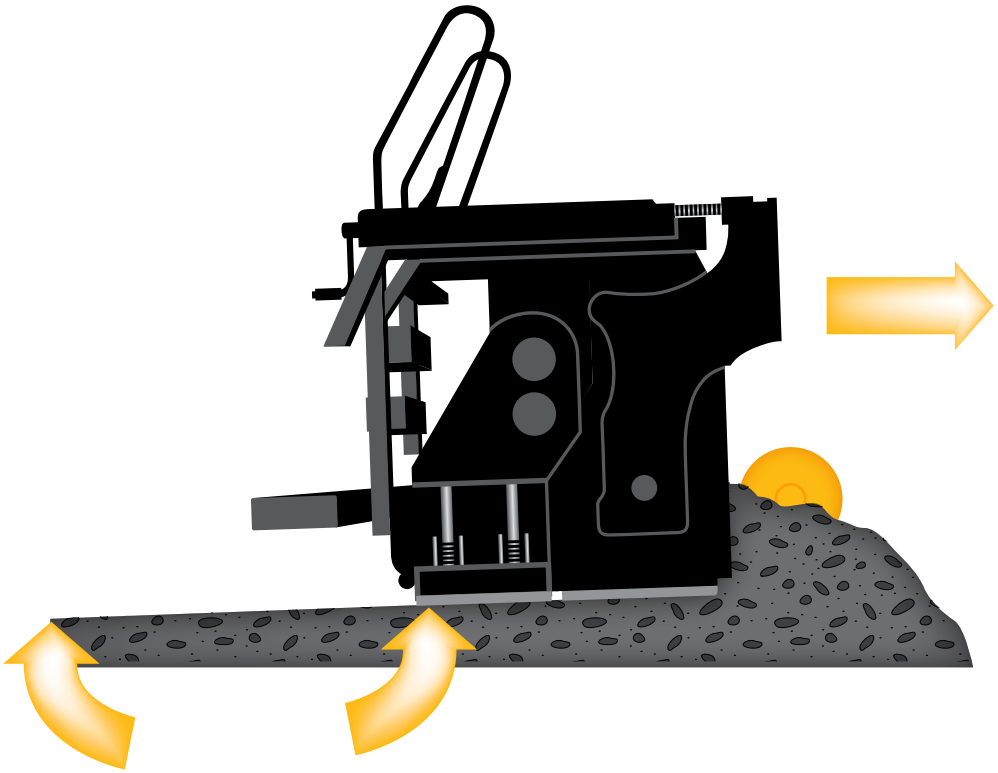


Note for tamper screeds: *If the paving speed increases and the tamper and vibration speeds remain constant, the distance between tamper impacts increases. Less material will be consolidated under the screed and the thickness will decrease. To maintain the correct layer thickness at a higher working speed, increase the tamper and vibration speeds accordingly.*

Decreasing the paving speed will cause the screed to climb.

When the paving speed is decreased, less energy is available to push the material in the auger chamber. The screed has a tendency to ride up the head of material and more mix passes under the screed.

Note for tamper screeds: *When working speed is decreased, impacts delivered by the tamper bar are closer together and more mix passes under the screed. To restore the balance between working speed and tamper forces, the speed of the tamper bar must be decreased.*



User Tip: *A common time for the operator to decrease the paving speed is when there is only one truck in front of the paver and no more in sight. Or, the operator may decrease the paving speed during truck exchanges to avoid running low on material in the hopper while continuing to pave. Short stops during truck exchanges or waiting for more material are not necessarily bad events. But, many operators have been trained to avoid, as much as possible, stopping the paver. Caterpillar recommends paving at a continuous, calculated speed with the understanding that short stops are often unavoidable.*

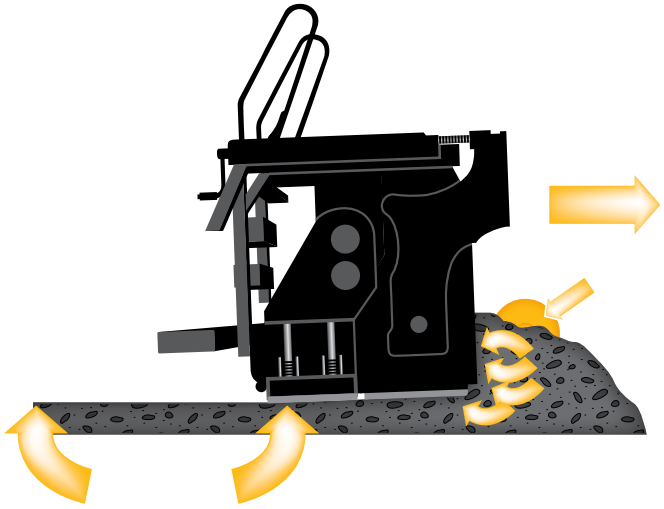
Material Feed System. Another factor that affects how the screed floats is the operation of the material feed system. The goal of the paving crew is to adjust the material feed system to create a continuous flow at uniform speed through the paver and in front of the screed.

The head of material in the auger chamber should cover one half, or slightly more than one half, of the augers.

The target for the paver operator and screed operators is to cover one half, to slightly more than one half, of the augers. Once the correct head of material is created, the crew must maintain it. Remember, the head of material in the auger chamber is the weight, or resistance, felt by the screed.

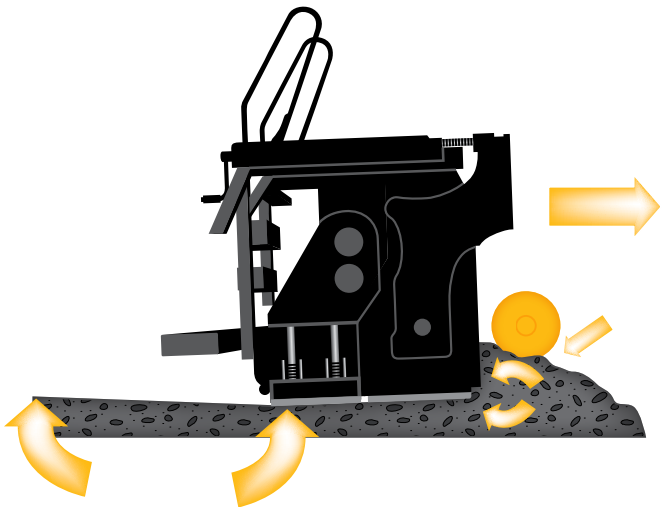
When the resistance to paving energy stays the same, then the screed stays at equilibrium and maintains its vertical position. Ride quality will be improved and the appearance of the bituminous layer enhanced.

Covering about half of the augers with material enables the mix to move efficiently toward the end gates at the full paving width. If the augers are overloaded, the material begins to compact in the auger chamber by its own weight and does not move as well. Auger overloading also requires more power and increases wear on the auger segments.



The screed height drops when the head of material decreases.

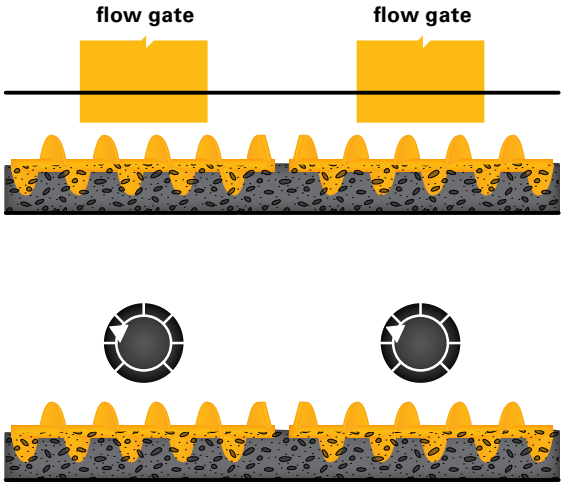
The screed will dive if the head of material in the auger chamber decreases. The loss of layer thickness happens very quickly and the screed may begin to drag aggregates. Many times the crew does not have to see this happening. They can hear the feeder system hydraulics begin to rapidly turn the augers as the feeder sensors detect the drop in material. Running low on mix in the auger chamber is considered a big mistake—one of the factors that has a big affect on quality.



User Tip: *There are several times that the head of material can diminish while paving. One is when paving under automatic slope control, because the screed will continue producing the correct slope at the surface of the layer. The screed may be laying a very thin layer or very thick layer if it is correcting for slope errors in the base. A very thick mat requires more material and can temporarily create a low head of material, until the feeder system catches up. The second possibility for running low on mix in the auger chamber is paving during truck exchanges. Some operators are trained to never stop between haul units. If the haul unit with a fresh supply of material is slow to position itself in front of the paver, the material in the hopper will run low, the conveyors will run low, and the head of material will run low.*

Controlling the head of material in the auger chamber and delivering a continuous flow of material through the paver requires coordination by the paver and the screed operators.

The paver operator has primary control of material delivery from the hopper through the tunnels and into the auger chamber. The material that drops off the conveyors into the auger chamber can be regulated in two ways, depending on the type of paver.



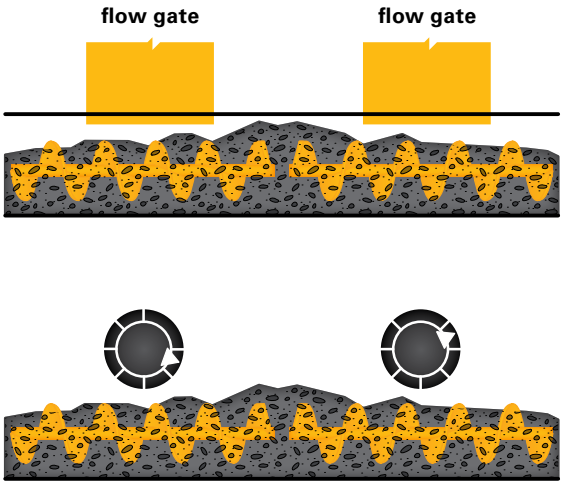
The position of the flow gates or the setting of the ratio control dials determines the level of material in the center of the auger chamber.

Some pavers have flow gates that are located over the left and right conveyor tunnels, on the rear wall of the hopper. Flow gates are strike-off devices used to meter the amount of material that flows back to the auger chamber. The height of each flow gate is independently adjustable.

Other pavers do not have flow gates, so the right and left conveyor tunnels are completely open. Those pavers have ratio control dials at the operator's station. Adjusting the ratio control dials regulates the speed of the conveyors.

Both flow gates and ratio control dials have the same affect on the material feed system. They control the volume of material that enters the auger chamber and, thus, they control the head of material in the center of the auger chamber.

If the operator raises the flow gates too high or turns up the ratio control dials too much, the auger chamber will be flooded and the head of material in the center of the auger will be too high. When there is too much material coming back on the conveyors, the rotational speed of the left and right augers will decrease. The augers may stop completely or run too slowly. The augers should not stop or operate in an ON / OFF mode during normal paving operations. Lower the flow gates or turn down the ratio control dials to speed up the rotation of the augers.

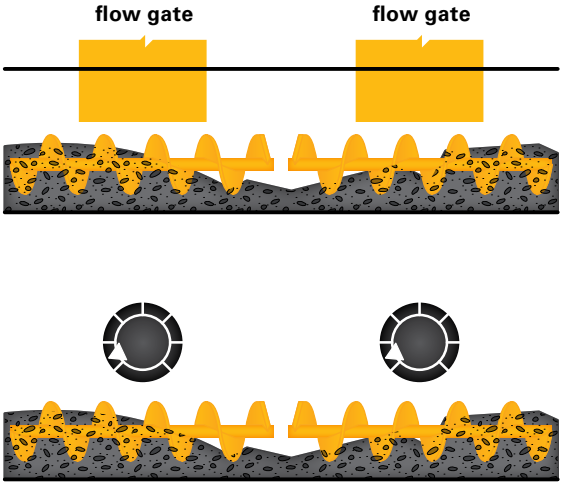


The head of material in the center of the auger will be too high if the flow gates are open too high, or the ratio control dials are set too high.

Conversely, if the flow gates are lowered too much or the ratio control dials are set too low, the center of the auger chamber will run low on mix and the head of material will be too low. When there is not enough material flowing back on the conveyors, the rotation of the augers will be too high.

The ideal range for auger speed is between 20 and 40 rotations per minute. If the auger speed is too high, continuous stripes of segregated aggregates may appear in the bituminous layer. If the auger speed is too slow or ON and OFF, random patches of segregated aggregates may appear. (More information about causes of stripe segregation and patch segregation are found in Unit 8.)

In addition to the appearance of various types of segregation, incorrect auger speed can also cause variations in the pre-compaction of mix in the auger chamber. Maintaining the correct auger speed takes teamwork on the part of the paver operator and the screed operator(s).



The head of material in the center of the auger will be too low if the flow gates are lowered too much, or the ratio control dials are set too low.

User Tip: *On most pavers, controls for flow gate height and conveyor speed (ratio control dials) are found at the operator's station on top of the paver. It is the paver operator who has the ultimate responsibility for controlling the volume of material coming into the auger chamber and, therefore, ultimate control of auger speed. The controls available for the screed operators can temporarily affect auger speed. The crew must understand this relationship.*

Screed operators control the head of material at the outer end of the auger shafts by adjusting the mix height dials at the screed control panels. To increase the head of material at the end of the auger shafts, turn the mix height dial clockwise. To decrease the head of material at the end of the auger shafts, turn the dial counter-clockwise. Adjusting the mix height dial affects the sensitivity of the material feed sensors that are aimed at the material emerging from the auger chamber.



Mix height dials at the screed control panels regulate the head of material at the ends of the auger shafts.

There are two types of feeder sensors: mechanical and sonic. Mechanical sensors contact the head of material as it flows through the auger chamber. As the material pushes against the paddle, the paddle turns a rheostat-type switch. As the paddle rotates outward, the sensor begins to slow the feeder system. As the paddle rotates inward, the sensor speeds the delivery of material. Adjusting the mix height dial controls where the sensor starts to slow the feeder system, or the point at which the system is shut off.



Mechanical feeder sensors contact the material as it flows out of the auger chamber.

User Tip: *Mixes that include modified asphalt cement are often sticky. Mix can build up on the tractor bulkhead and can interfere with the motion of a mechanical feeder sensor. The crew needs to periodically clean the tractor bulkhead to avoid erratic feeder system operation.*

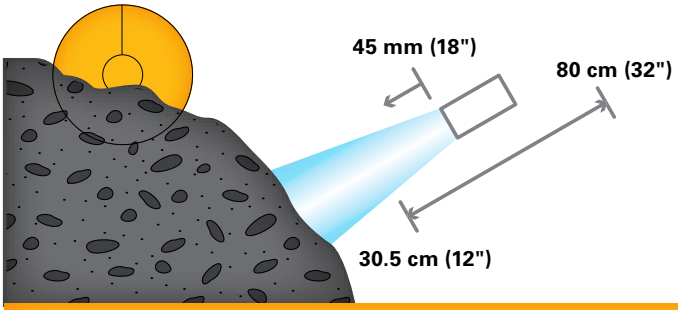
The other type of feeder sensor is a sonic-type sensor. This sensor emits sound pulses. Echoes bounce off the material coming out of the auger chamber. The sonic sensor monitors the time it takes for the echoes to return to measure the distance between the sensor and the target. There are several principles that are important to know about sonic feeder sensors.

The working range for a sonic feeder sensor is between 30-80 cm (12-32"). If the target is closer than 30 cm (12"), the feeder system for that side of the paver will shut off. If the target for a sonic feeder sensor is beyond 80 cm (32"), then the feeder system for that side of the paver will run at full speed. The recommended distance from the target for sonic feeder sensors is 45 cm (18").



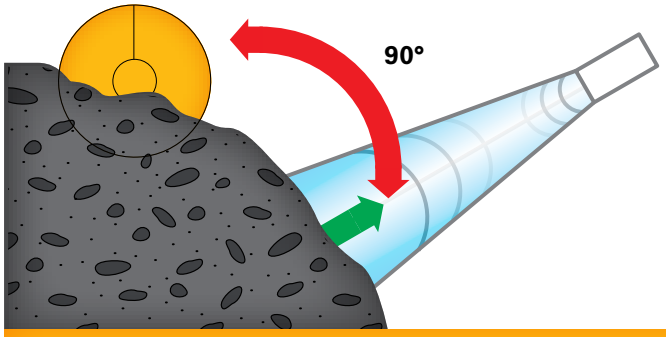
Sonic feeder sensors emit sound pulses that rebound off the material coming out the auger chamber and back to the sensor.

Aiming a sonic feeder sensor is extremely important. It emits a given number of sound pulses per second. In order to be accurate as a measurement device, the sensor must receive the same number of returning echoes. If the sensor is aimed at an incorrect angle, some of the echoes will not return to the sensor and the feeder system will operate erratically.



Sonic feeder sensors must be positioned the correct distance away from the intended target.

User Tip: *If difficulty is encountered getting the feeder system to run smoothly at the desired speed and desired head of material, the first troubleshooting step should be to re-aim the sonic sensor until the system operates properly. As an aid to aiming the sensor, use a tape measure or folding ruler extended to 45 cm (18"). Hold the ruler on the end of the sensor and see what the ruler is contacting at that distance. Check the angle and make sure that the sensor is pointed directly at the target.*



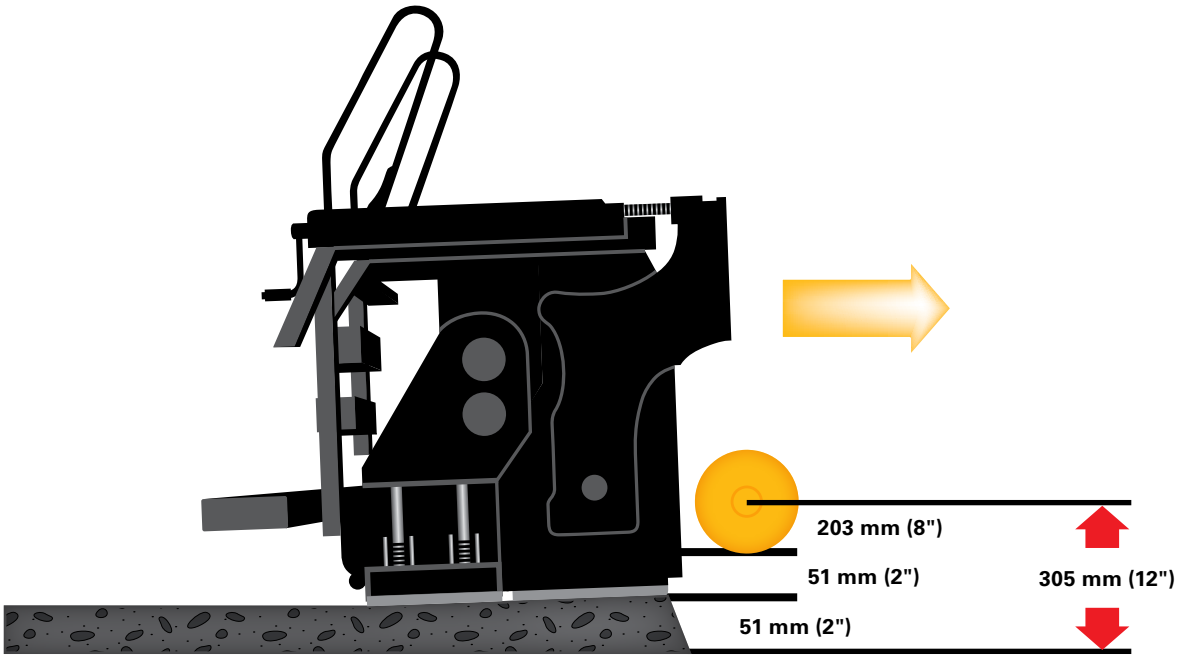
Sonic feeder sensors must be aimed perpendicular to the intended target.

Selecting the best target is another key to smooth feeder system operation when using sonic feeder sensors. Ideally, the sensor should be aimed at the material coming off the front side of the auger shaft. This material is moving toward the screed end gate, which is the outer edge of the bituminous layer. If this material is targeted, the mix height dial will help control how the material flows out and where it contacts the end gate. Do not aim the sensor at the grade or too far back on material that is stagnant. Always aim the sensor at moving material.

The target for the sonic sensor should be the active head of material coming off the front side of the auger shaft.



User Tip: *One of the most difficult applications when using a sonic feeder sensor is paving at a width that is the same as the basic screed width. In other words, the end gate at the screed extender is drawn all the way in, and no material is flowing from the auger chamber. In that situation, the sensor will have to be aimed straight down. Try to keep the sensor at least 30 cm (12") away from the target. Or, if mechanical sensors are available, install them. Mechanical sensors make it easier to control the feeder system at narrow paving widths.*



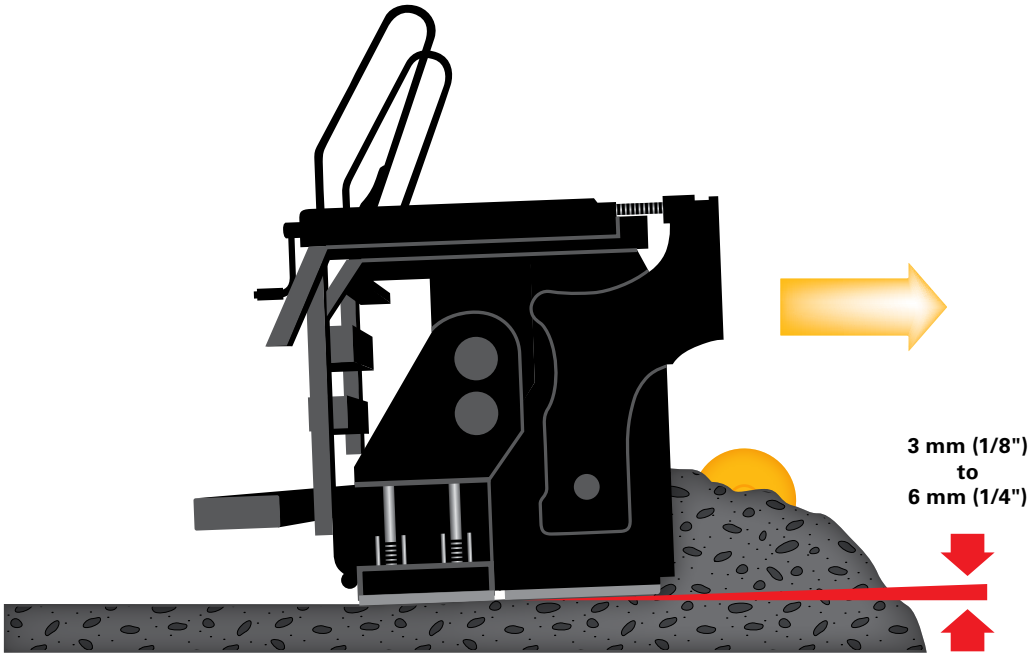
Adjust auger height to avoid creating auger shadows in the surface of the bituminous layer.

On many pavers, the height of the auger shaft is adjustable. When preparing the paver for operation, lower the augers until they are about 5 cm (2") above the surface of the layer for mixes with a maximum aggregate size of 13 mm (0.5") or smaller. For mix designs with aggregates 19 mm (0.75") or larger, position the augers 8 cm (3") above the layer.

If the augers are too low, they will leave dark shadows in the surface of the bituminous layer. The layer will appear tight in the center and will have open texture (shadows) on either side of center, the same length as the left and right auger shafts. Then, the layer will appear tight again at the edges. If this texture pattern develops, gradually raise the augers until the appearance of the layer is uniform.

User Tip: *Here's an easy way to measure auger height. Assume that the augers are 40 cm (16") in diameter. The center of the auger shaft will be 20 cm (8") above the bottom of the augers. The bottom of the augers should be 5 cm (2") above the layer. Assume that the layer is 5 cm (2") thick. Lower the augers before filling the auger chamber. Measure from the center of the auger shaft to the grade. When the distance is 30 cm (12"), the auger height is correct for that layer thickness. Add 2.5 cm (1") to auger height when using a large-stone mix.*

Screed Adjustments – The third crew-controlled factor that affects how the screed floats is screed setup and adjustment. The first setup step is establishing the correct angle of attack.



At equilibrium, the screed should float at a slight nose-up attitude.

All screeds are designed to produce the best results when the screed reaches equilibrium and floats at a slight nose-up attitude. The front of the screed plate should be between 3-6 mm (0.125-0.25") higher than the trailing edge of the screed plate. That angle of attack (also called the planing angle) allows material to pass under the nose of the screed and provide the support that floats the screed. At the correct angle of attack, the entire surface of the screed plate is used to help provide pre-compaction and the surface texture of the bituminous layer.



Texture stripes in the surface of the bituminous layer are usually caused by variations in the angle of attack across the width of the screed.

When the angle of attack for the main screed plate or an extender screed plate is too high, a smaller portion of the screed is in contact with the bituminous layer. The smaller contact area exerts more force on the surface of the layer and will create a tight, shiny appearance at the surface.

When the angle of attack is too low or when the screed is running nose down, the surface looks open or torn, and short waves can be seen. Note that the surface of the layer in the photo is open behind the left extension (right side of photo) and tighter behind the main screed. If the density is checked at different locations across the width of the layer, you will find that the pre-compaction density will be variable, too.



The angle of attack on screed extensions should be adjusted while the paver is moving.

The angle of attack at the main screed extenders are set at the start of paving and is adjusted to change the layer depth. The angle of attack at the screed extensions is fixed when screed plates are installed. When paving starts, it is common to adjust the extender angle of attack to make the surface texture uniform.

On many screeds, there is a series of adjusters on the trailing edges of the extenders. Turn the adjusters clockwise to increase the angle of attack. Per the example shown in the photo above, the screed operator is gradually increasing the angle of attack on the left extender to tighten the surface of the layer.



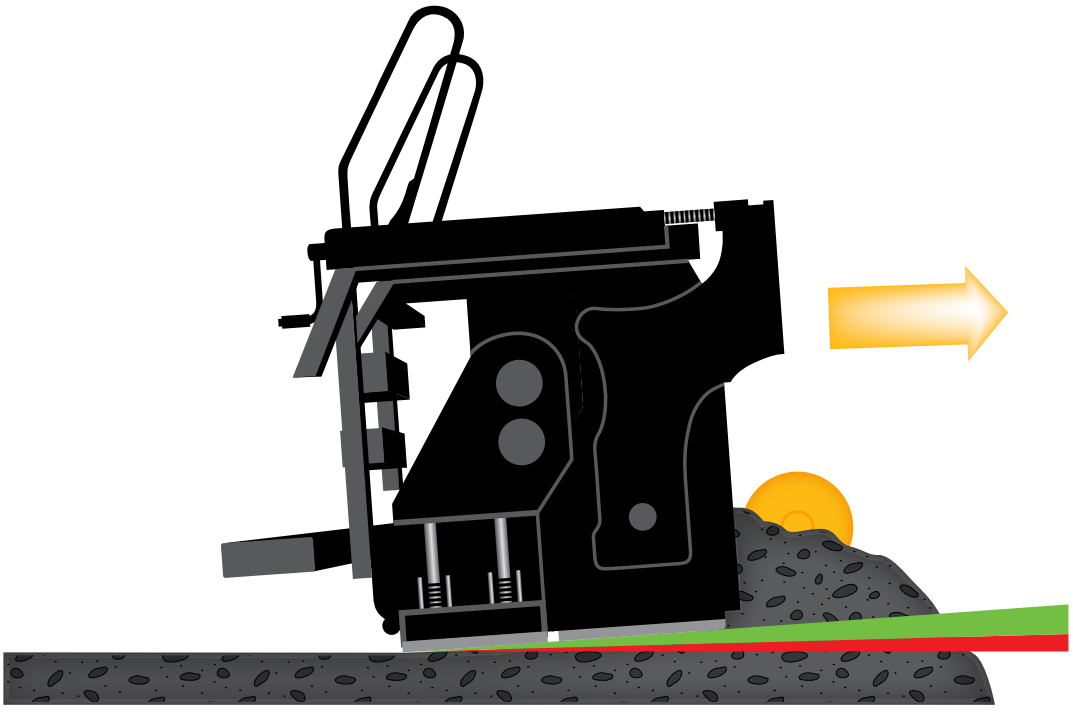
When the planing angles are the same on the main screed and the screed extenders, the texture of the surface is uniform.

The surface of the bituminous layer should have a uniform appearance from edge to edge. Remember, texture is affected by angle of attack. Material segregation and other defects have other causes. (For more information about defects and problems, see Unit 8.)

User Tip: *There are several other ways to determine if the screed is consistently running with the wrong angle of attack. If the angle is too high, there will be excessive and premature wear on the trailing edge of the screed plate when compared to the nose bar and the front of the screed plate. If the angle of attack is too low, or if the screed is running nose down, the wear will be concentrated around the nose bar. Also, lock the screed when raising it at the end of a pass, and look at the bottom of the plates. The screed was tilted back and running with too much angle of attack during that pass if the screed plates are clean and shiny at the rear half, and have fines adhering to the front half.*

In review, a few noteworthy principles affect the angle of attack on the main screed.

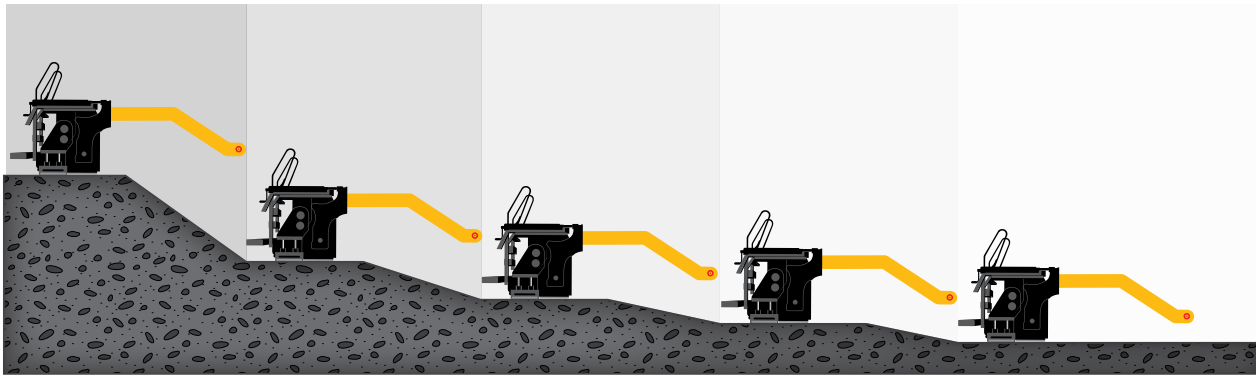
Every time the screed is lowered on a starting reference and prepared for paving, adjustments must be made that create the angle of attack for the main screed. Those adjustments are covered in detail in Units 4 and 5. For starters, assume that the main screed has the correct angle of attack—3-6 mm (0.125-0.25").



While paving, the screed operator can increase or decrease the angle of attack.

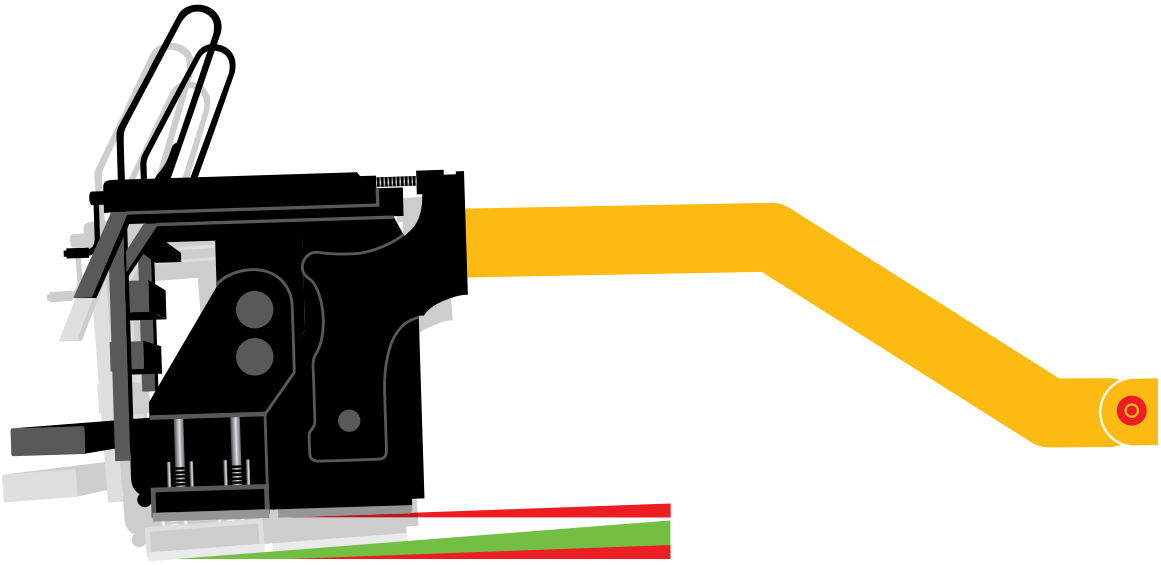
There are two ways a screed operator can change the angle of attack while paving. On a vibratory screed, the operator can turn the depth-control screws located on the left and right sides of the screed. Or, the operator can adjust the tow-point height up or down manually, or by using automatic grade and slope control. On a tamper bar screed, the operator normally uses only the tow-point height as a means to change the main screed angle of attack.

User Tip: *If the paver is equipped with depth control screws and those screws have conventional thread spacing, one rotation of the screw will cause about 6 mm (0.25") of screed movement. If the depth control screws have coarse threads, one rotation of the screw will cause about 13 mm (0.5") of screed movement.*



The change in layer thickness occurs over a distance equal to five tow arm lengths.

When a depth-control screw is turned or when the tow-point height is adjusted, the angle of attack changes and the screed immediately begins to move up or down. Most of the change in layer thickness, about 65 percent, occurs in the first tow-arm length. Another 65 percent of the remainder occurs in the next tow-arm length. The thickness change occurs progressively until the change is complete in five tow lengths. This phenomenon is the leveling effect that is common to all floating screeds. The gradual change in layer thickness improves the smoothness of the layer.

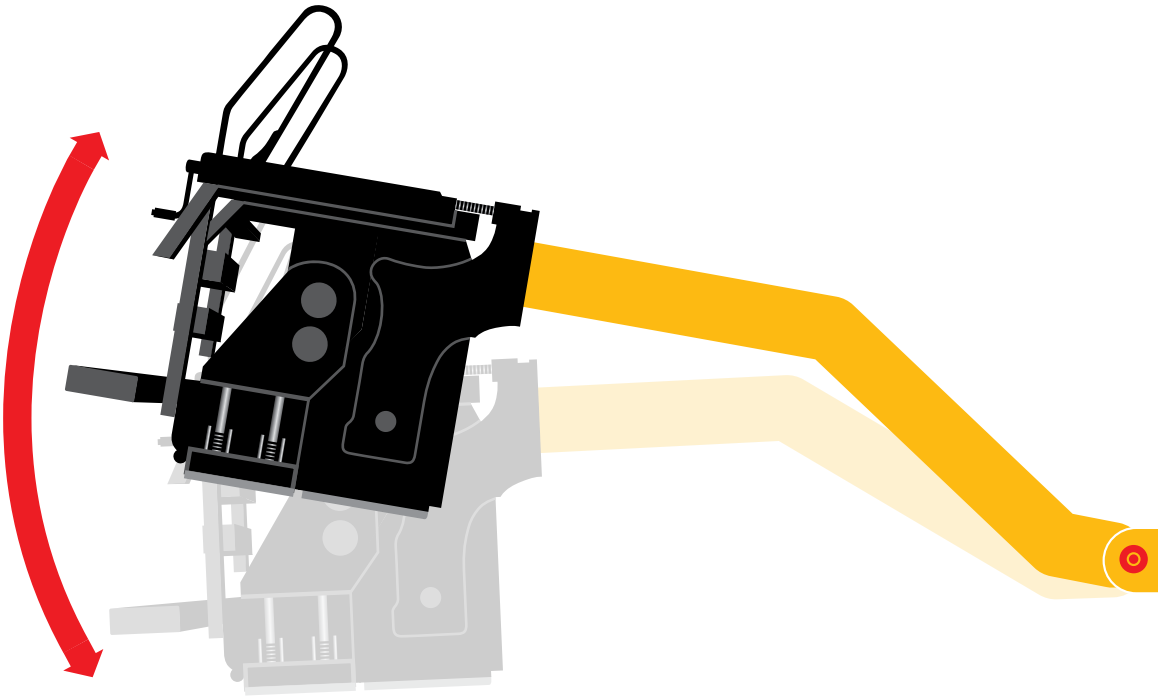


When the thickness change is complete, the screed has returned to its original angle of attack.

The thickness change is complete at the end of five tow-arm lengths. The main screed has returned to its original angle of attack and resumed its equilibrium point where it is floating at an unchanging vertical position.

The screed operator should wait until the screed has resumed its original angle of attack (reached equilibrium) before checking layer thickness. If the screed has depth-control screws, the operator can check the tension on the depth-control screw.

When the thickness change has completely occurred, the depth-control screw should have no tension—indicative of the floating screed. If there is still tension on the depth-control screw, the screed is still moving and has not returned to the original angle of attack.



The screed moves in an arc as it changes layer thickness.

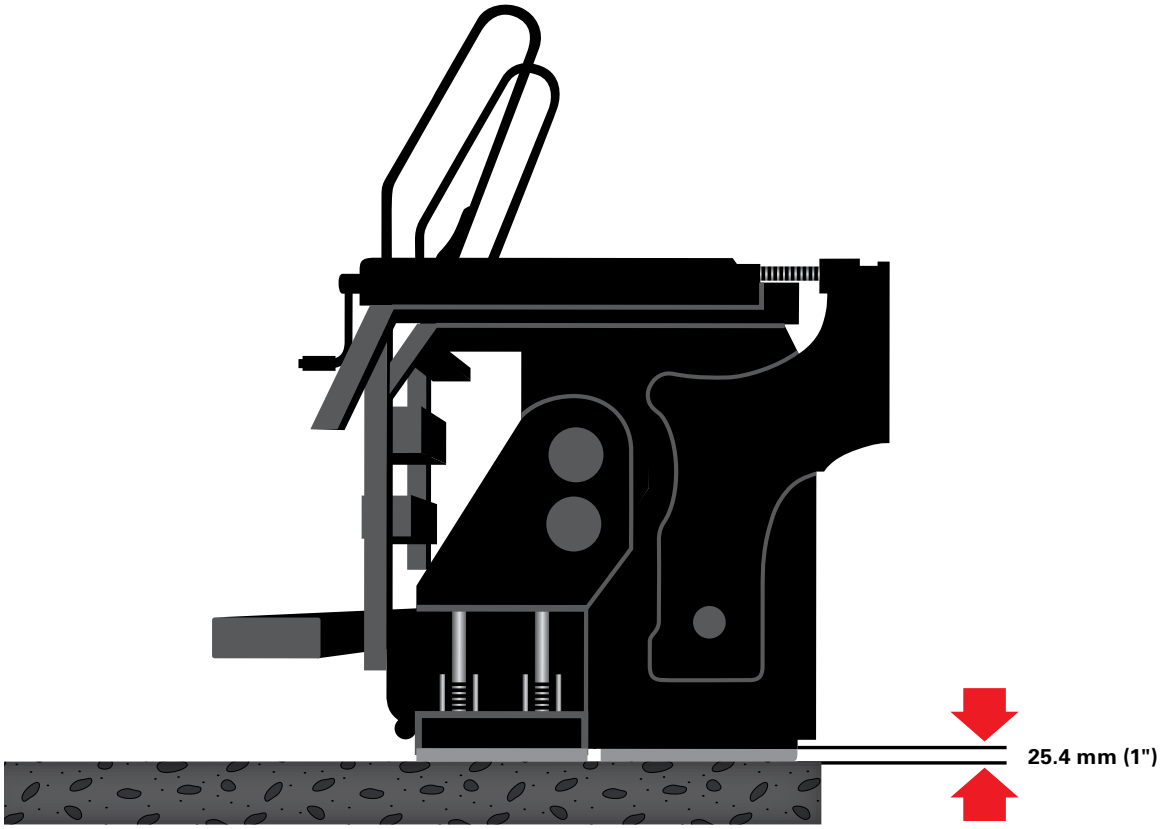
Understanding how the screed moves up and down is key to understanding why the screed always returns to the angle of attack that the crew created at the start of paving.

Remember, the screed is attached to the paver on both sides at the tow-arm connections. Think of that connection as the center of a circle. The length of the tow arm where it attaches to the front of the screed is a radius. The length of the tow arm plus the screed itself forms a longer radius. Since the tow arm is pinned to the tow-point connection, the screed actually rotates around that pin connection as it moves.

The front of the screed, which has a shorter radius, moves in a smaller arc, not straight up and down. The rear of the screed, which has a longer radius, moves in a larger arc compared to the screed nose. Therefore, the rear of the screed moves a little farther and a little faster than the nose of the screed. The change in angle of attack at the front of the screed is gradually canceled by the fact that the rear of the screed is moving to return the screed to equilibrium—the original angle of attack.

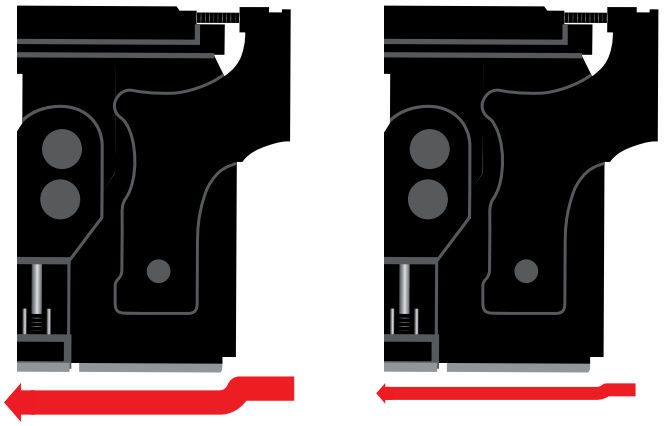
Units 4 and 5,contain information about the procedure to set the correct angle of attack at the start of paving. Correctly setting the angle of attack at the start is critical, because the screed always returns to that position after every thickness correction.

Several other screed adjustments should also be considered.



Adjustable strike-offs on some screeds control the exposure of the nose bar.

Some vibratory screeds have an adjustable strike-off on the front wall of the main screed. The strike-off height controls the amount of the rounded nose bar that is exposed. The factory height is 25 mm (1") above the screed plate, so the entire nose bar is exposed. For many mix designs, the nose bar should be fully exposed to aid material flow.



Lowering the strike-off reduces the flow of material under the nose bar.

When the strike-off is lowered, it's harder for material to flow under the rounded nose bar. Caterpillar recommends lowering the strike-off to counteract the effects of paving with mixes that contain large aggregate sizes.

Some base layer mixes have aggregates up to 38 mm (1.5"). Those large aggregates tend to float the screed easily and create too much lift. It becomes difficult to control layer thickness without reducing the angle of attack too much. The screed runs flat or nose down. The surface or the bituminous layer looks open, torn or scuffed because the angle of attack is incorrect. Closing off some of the nose bar reduces the lifting forces exerted by mixes containing large aggregates and helps the screed float at the correct angle of attack.

The screed counterbalance system is another feature that affects screed performance. When the counterbalance system is activated, it exerts a small amount of upward hydraulic pressure (lift) through the screed lift cylinders. Its purpose is to help reduce screed settlement marks when the paver is stopped. It is most useful when working with sandy or tender mixes.



Screed counterbalance is a standard feature on Cat Pavers.

On newer Cat Pavers with vibratory screeds, counterbalance pressure can be viewed and adjusted via the Advisor display. Older Cat Pavers or Cat Pavers with tamping screeds are equipped with a pressure gauge and adjustment dial.



When the paver stops on the fresh bituminous material, it may leave an impression of the screed in the surface of the layer.

The screed will dent the fresh bituminous material to a certain extent—depending on the mix type, the length of the paver stop, and the screed weight. The marking will be most apparent when paving with tender mixes, which have smaller aggregates, and straight-run asphalt cement. The mark will be less apparent when paving with stiff mixes, especially those with modified asphalt cement.



Examine the surface of the layer after the initial-phase compactor has completed a pass over the screed indentation.

After paving starts again, wait for the initial-phase compactor to make a pass over the area where the paver stopped. The screed mark should no longer be visible. In other words, the compaction process will completely eliminate the marks and they will not affect ride quality. If the marks remain visible after the initial-phase compactor has completed its passes, activate the screed counterbalance system or, if the system is already in use, increase the pressure.

On pavers with vibratory screeds, the factory setting is 3.4 bar (50 psi). If counterbalance is required, reduce the pressure to the lowest possible setting to achieve the desired results, never to exceed 10.3 bar (150 psi).

On pavers with tamping screeds, set the system pressure between 0 and 5 bar (0 and 73 psi) for the Paving mode and 15 bar (218 psi) for the Standby mode. Adjust the Standby pressure as required to eliminate deep screed settlement marks.

Cat Pavers with tamping screeds are equipped with a screed lower lock function.

When screed lower lock is engaged, the screed is put in the Hold position when the paver controls are in the Standby mode. Therefore, the screed is virtually weightless when the paver is stopped, which minimizes indentations. When paving resumes, the Standby mode is terminated and the screed lower lock disengages after a preset time.

[SUMMARY]

Understanding how a paver works and what factors impact the concept of the free-floating screed is critical to achieving high quality results when laying down bituminous material. Each crewmember should know the theory that is behind the process, not just the steps in the paving process. If every member of the crew, including supervisors and quality control personnel, can explain theory, they will be better equipped to troubleshoot problems when they occur.

Caterpillar recommends continuous training that includes a review of fundamentals and principles of paving. Even the most experienced operators and supervisors will benefit from exposure to basic training. A thorough knowledge of the basics lays the foundation for advanced study.


AP655F
CAT
GRADE CONTROL



Unit 4
PREPARING TO PAVE WITH VIBRATORY SCREEDS

Among the most crucial steps during takeoff are setting the angle of attack; filling the auger chamber slowly, one side at a time; and paving with a calculated speed.



Before pulling off the starting joint, the crew must heat the screed plates to prevent hot mix from sticking to them.

Unit 4 describes how to prepare a paver to pull off a starting reference when utilizing a vibratory screed. The process for getting ready to pave with a tamping and vibrating screed is different and will be covered in Unit 5. To prepare the paver, Caterpillar recommends using a step-by-step procedure that helps crewmembers work together through all of the required steps.

Pulling away from a starting reference can be a frustrating process for even the most experienced paving crew. Sometimes the screed drops after pulling off the starting reference. Sometimes the screed climbs during the takeoff. Sometimes the takeoff is perfect. While there is no way to guarantee perfect takeoffs every time, the crew can create more predictable results when they work together to follow a logical sequence.

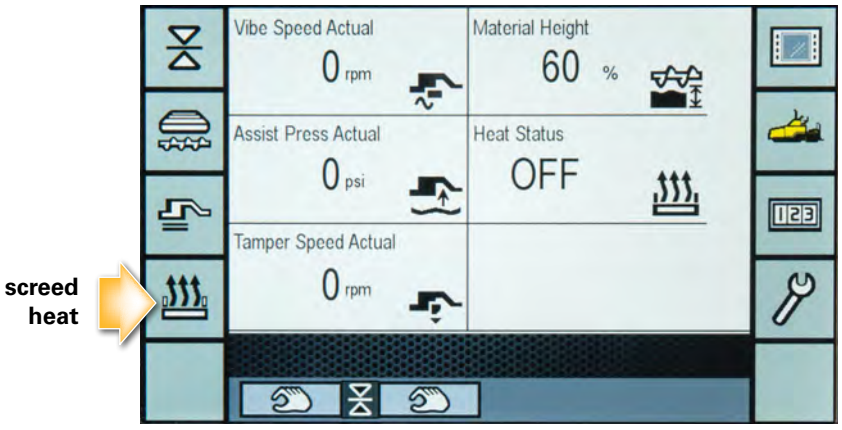
Note: *Some of the steps described in this unit apply specifically to certain Cat Pavers equipped with vibratory screeds. Some procedures may be different for older Cat Paver models. Other steps are general in their application and are appropriate for any paver model. Always consult the specific instructions in the Operation and Maintenance Manual for the pavers used on the worksite.*

[1. HEAT THE SCREED]

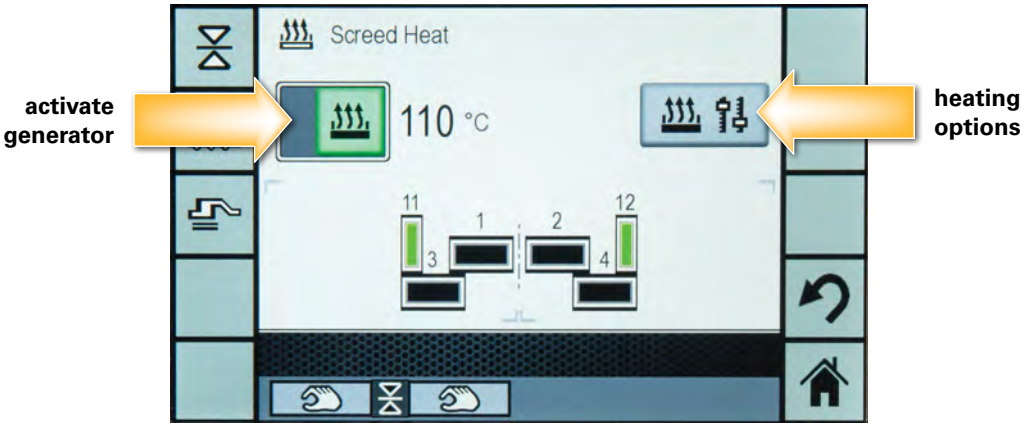
The first step is heating the screed plates prior to pulling off the starting reference. If the screed plates are not close to the temperature of the material passing under the screed, the hot mix will adhere to the bottom of the screed plates. Drag marks will appear in the surface of the bituminous layer, and the screed will dive after pulling off the starting reference.

Before heating the screed, position the screed approximately 25-75 mm (1-3") off the ground. If the screed is in the fully raised position, wind can affect the performance of the heating system. If the screed is in the fully lowered position, the ground temperature affects the performance of the heating system. In both cases, the heating system will take longer to bring the screed plates up to the desired temperature.

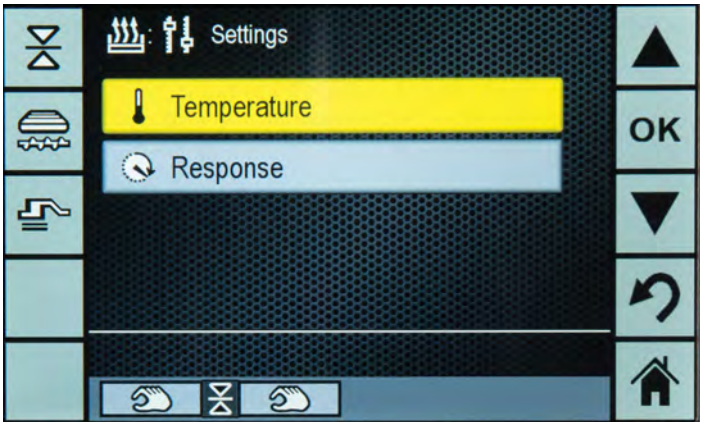
Two types of screed heating systems are available on Cat Pavers. One is electric screed heat.



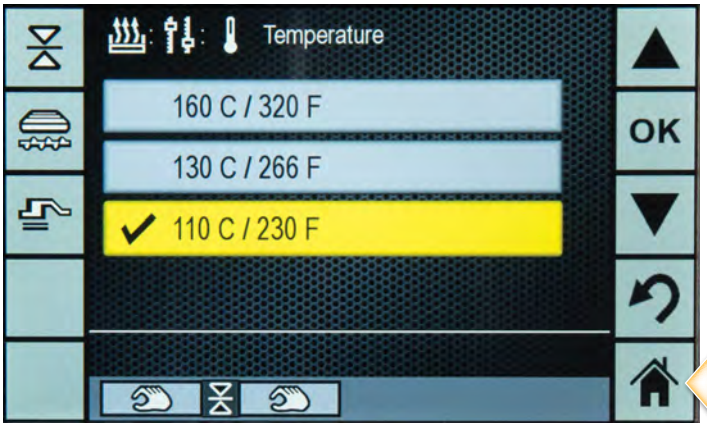
Use the display at either tractor operating station or either screed operating station to start the screed heating system. To start electric screed heat, first touch the screed heat icon on the display screen.



The screed-heat screen will appear. The generator icon is located on the left side. Touch the generator icon and slide to the right. When activated, the icon will turn green, the engine rpm will increase to 1300 rpm, and the generator will be activated. Choose the temperature set-point by selecting the heating option icon on the right side of the screen.



The heating options menu appears next in the display. Touch the temperature option.



The screen then shows three choices for set-point temperature. Select either low range, where the set-point temperature will be 110° C (230° F); the mid range, where the set-point temperature will be 130° C (266° F); or the high range, where the set-point temperature will be 160° C (320° F). Once the temperature range has been selected, touch the Home screen.